

```
PES1UG24CS320:~$ mkdir rootfs-base
wget https://dl-cdn.alpinelinux.org/alpine/v3.20/releases/x86_64/alpine-minirootfs-3.20.3-x86_64.tar.gz
tar -xzf alpine-minirootfs-3.20.3-x86_64.tar.gz -C rootfs-base
```

```
cp -a rootfs-base rootfs-alpha
cp -a rootfs-base rootfs-beta
mkdir: cannot create directory 'rootfs-base': File exists
--2026-04-21 17:23:35-- https://dl-cdn.alpinelinux.org/alpine/v3.20/releases/x86_64/alpine-minirootfs-3.20.3-x86_64.tar.gz
Resolving dl-cdn.alpinelinux.org (dl-cdn.alpinelinux.org)... 151.101.158.132, 151.101.158.133, 151.101.158.134, 151.101.158.135
Connecting to dl-cdn.alpinelinux.org (dl-cdn.alpinelinux.org)|151.101.158.132|:443
HTTP request sent, awaiting response... 200 OK
Length: 3490290 (3.3M) [application/octet-stream]
Saving to: 'alpine-minirootfs-3.20.3-x86_64.tar.gz'
```

```
alpine-minirootfs 100%[=====>] 3.33M 6.41MB/s in 0.5s
```

```
2026-04-21 17:23:36 (6.41 MB/s) - 'alpine-minirootfs-3.20.3-x86_64.tar.gz' saved
```

```

[sharvari@PES1UG24CS320 ~]$ sudo ./engine supervisor rootfs-alpha/
[sudo] password for sharvari_naik:
sudo: ./engine: command not found
PES1UG24CS320:~$ cd ~/os-jack
PES1UG24CS320:~/os-jack$ sudo ./engine supervisor rootfs-alpha/
✓ Supervisor started with rootfs: rootfs-alpha/
Waiting for commands...

```

```

PES1UG24CS320:~/os-jack$ ./engine start c1 rootfs-alpha/ "sleep 40"
Starting container c1...
Container started with PID: 7383
PES1UG24CS320:~/os-jack$

```

```
Starting container c1...
Container started with PID: 7383
PES1UG24CS320:~/os-jack$ ps -o pid,ni,cmd | grep sleep
 7383    0 sh -c sleep 40
 7384    0 sleep 40
 7405    0 grep --color=auto sleep
PES1UG24CS320:~/os-jack$ █
```

```
 7405    0 grep --color=auto sleep
PES1UG24CS320:~/os-jack$ ./engine stop c1
ps -o pid,ni,cmd | grep sleep
Stopping container...
 7383    0 sh -c sleep 40
 7384    0 sleep 40
 7415    0 grep --color=auto sleep
PES1UG24CS320:~/os-jack$ █
```

```
7428 0 grep --color=auto sleep
PES1UG24CS320:~/os-jack$ ./engine start c1 rootfs-alpha/ "sleep 40"
./engine start c2 rootfs-beta/ "nice -n 10 sleep 40"
Starting container c1...
Container started with PID: 7428
Starting container c2...
Container started with PID: 7431
PES1UG24CS320:~/os-jack$
```

```
PES1UG24CS320:~/os-jack$ ps -o pid,ni,cmd | grep sleep
7428 0 sh -c sleep 40
7430 0 sleep 40
7431 0 sh -c nice -n 10 sleep 40
7432 10 sleep 40
7458 0 grep --color=auto sleep
PES1UG24CS320:~/os-jack$
```

```
PES1UG24CS320:~/os-jack$ ./engine start c1 rootfs-alpha/ "sleep 40"
./engine start c2 rootfs-beta/ "nice -n 10 sleep 40"
Starting container c1...
Container started with PID: 7428
Starting container c2...
Container started with PID: 7431
PES1UG24CS320:~/os-jack$ ps -o pid,ni,cmd | grep sleep
7428 0 sh -c sleep 40
7430 0 sleep 40
7431 0 sh -c nice -n 10 sleep 40
7432 10 sleep 40
7458 0 grep --color=auto sleep
PES1UG24CS320:~/os-jack$ ./engine ps
Listing containers...
  PID TTY          TIME CMD
 7359 pts/3        00:00:00 bash
 7468 pts/3        00:00:00 engine
 7469 pts/3        00:00:00 sh
 7470 pts/3        00:00:00 ps
PES1UG24CS320:~/os-jack$
```

```

7458  0 grep --color=auto sleep
PES1UG24CS320:~/os-jack$ ./engine ps
Listing containers...
  PID TTY          TIME CMD
 7359 pts/3        00:00:00 bash
 7468 pts/3        00:00:00 engine
 7469 pts/3        00:00:00 sh
 7470 pts/3        00:00:00 ps
PES1UG24CS320:~/os-jack$ ./engine logs c1
Fetching logs...
PES1UG24CS320:~/os-jack$ █

```

```

PES1UG24CS320:~/os-jack$ ./engine stop c1
Stopping container...
PES1UG24CS320:~/os-jack$ █
PES1UG24CS320:~/os-jack$ ps -o pid,ni,cmd | grep sleep
 7543  0 grep --color=auto sleep
PES1UG24CS320:~/os-jack$ ./engine start c3 rootfs-alpha/ "yes > /
dev/null"
Starting container c3...
Container started with PID: 7564
PES1UG24CS320:~/os-jack$ █

```

```

top - 17:30:36 up 57 min,  2 users,  load average: 3.25, 3.06, 2.
Tasks: 295 total,   4 running, 291 sleeping,   0 stopped,   0 zom
%Cpu(s): 89.6 us, 10.4 sy,   0.0 ni,   0.0 id,   0.0 wa,   0.0 hi,   0
MiB Mem :  3875.8 total,   715.0 free,  2005.8 used,  1155.0
MiB Swap:   0.0 total,   0.0 free,   0.0 used.  1956.6

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM
7565	sharvar+	20	0	5768	1004	916	R	61.0	0.0
6082	sharvar+	20	0	5768	1020	932	R	57.7	0.0
6353	sharvar+	20	0	5768	1012	924	R	50.0	0.0
5381	sharvar+	20	0	11.8g	658212	231596	S	9.0	16.6
5759	sharvar+	20	0	3338596	602760	108968	S	8.7	15.2
1987	sharvar+	20	0	4129828	191932	69156	S	7.1	4.8
2318	sharvar+	20	0	567424	41768	23380	S	1.6	1.1
5066	root	20	0	0	0	0	I	1.0	0.0
7582	sharvar+	20	0	10616	4188	3320	R	1.0	0.1
6914	root	20	0	0	0	0	I	0.6	0.0
832	root	20	0	241076	7176	5728	S	0.3	0.2
1110	root	10	-10	309792	47532	12520	S	0.3	1.2
1539	root	20	0	295620	8052	5136	S	0.3	0.2
2120	sharvar+	20	0	459992	8276	3268	S	0.3	0.2

```

2318 sharvar+ 20 0 567424 41768 23380 S 0.3 1.1
5604 sharvar+ 20 0 450404 37768 27732 S 0.3 1.0
5623 sharvar+ 20 0 2647748 157256 87908 S 0.3 4.0
PES1UG24CS320:~/os-jack$ ./engine stop c3
Stopping container...
PES1UG24CS320:~/os-jack$ █

```

```

PES1UG24CS320:~/os-jack$ ./engine stop c3
Stopping container...
PES1UG24CS320:~/os-jack$ ./engine start c4 rootfs-alpha/ "sleep 5
"
Starting container c4...
Container started with PID: 7605
PES1UG24CS320:~/os-jack$
PES1UG24CS320:~/os-jack$ ./engine start c4 rootfs-alpha/ "sleep 5

```

```

Starting container c4...
Container started with PID: 7605
PES1UG24CS320:~/os-jack$ ps aux | grep Z
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIM
COMMAND
sharvar+      7623  0.0  0.0   6480  2268 pts/3    S+   17:31   0:0
grep --color=auto Z
PES1UG24CS320:~/os-jack$ ./engine start c5 rootfs-alpha/ "echo he
lo"
Starting container c5...
Container started with PID: 7636
PES1UG24CS320:~/os-jack$ hello

```

```

make[1]: Leaving directory '/usr/src/linux-headers-5.15.0-174-generic'
PES1UG24CS320:~/os-jack$ sudo insmod monitor.ko
[sudo] password for sharvari_naik:
PES1UG24CS320:~/os-jack$ ls /dev/container_monitor
/dev/container_monitor
PES1UG24CS320:~/os-jack$ gcc engine.c -o engine -lpthread
PES1UG24CS320:~/os-jack$ mkdir rootfs
PES1UG24CS320:~/os-jack$ cd rootfs
PES1UG24CS320:~/os-jack/rootfs$ mkdir bin lib lib64 proc
PES1UG24CS320:~/os-jack/rootfs$ cd ..
PES1UG24CS320:~/os-jack$ sudo ./engine supervisor rootfs/
Unknown command
PES1UG24CS320:~/os-jack$ gedit engine.c
PES1UG24CS320:~/os-jack$ gcc engine.c -o engine -lpthread
PES1UG24CS320:~/os-jack$ sudo ./engine supervisor rootfs/
✓ Supervisor started with rootfs: rootfs/
Waiting for commands...

```